

Stability of mechanical systems subjected to force fields.

1 Introduction

Until now in this course, we have assumed the forces acting on the system to be explicitly known as function of time: this is the case e.g. for the harmonic force generated by a rotating unbalanced mass or for a periodic force. In other cases, however, the force may depend on the state of the system. We will denote this case as the system being subjected to a *force field*.

We assume in this lesson the mechanical system to be linear, but the force field to be non-linearly depending on the state of the system. The equation of motion for a single-degree of freedom (d.o.f.) system subject to a field force takes then the form:

$$m_s \ddot{x} + c_s \dot{x} + k_s x = F(x, \dot{x}, t) \quad (1.1)$$

where m_s , c_s and k_s are respectively the mass, damping and stiffness parameters of the system (subscript s is used here to denote the *structural* parameter of the system in contrast to the additional damping and stiffness added by the force field, see Section 3) and F is the state-dependent generalised force acting on the system due to the effect of the force field.

For a multi-d.o.f. system, the equations of motion take the matrix form:

$$[m_s] \ddot{\underline{x}} + [c_s] \dot{\underline{x}} + [k_s] \underline{x} = \underline{F}(\underline{x}, \dot{\underline{x}}, t) \quad (1.2)$$

with $[m_s]$, $[c_s]$ and $[k_s]$ the structural mass, damping and stiffness matrices of the system.

In the case when F does not depend explicitly on time, we say the force field is *autonomous*, otherwise we say it is *non-autonomous*. In this lesson, we are only concerned with the study of autonomous systems.

Because a field force depends on the state of the system, it will affect the vibrating properties of the system (namely, the natural frequencies and modes of vibration), and it may also affect the stability of the system.

In this lesson we will indeed focus on the stability of a mechanical system subjected to a force field. First, we will recall some techniques for analysing the stability of a non-linear dynamic system in the neighbourhood of an equilibrium configuration (Section 2). Then we will perform a stability analysis for a single-d.o.f. system (Section 3) and for a 2-d.o.f. system (Section 3).

2 Stability of a dynamic system in the neighbourhood of an equilibrium position.

The stability of a dynamic system is concerned with how the system reacts to a perturbation, i.e. an initial condition (e.g. caused by a transient disturbance) that deviates the system from its previous motion, called the *unperturbed motion*. The motion performed by the system following the perturbation is termed the *perturbed motion*.

In simple terms, the system is said to be stable if, for any perturbation (possibly with limitations on its amplitude, see below in this section) the perturbed motion of the system will remain sufficiently ‘close’ to the unperturbed one. However, there is a need to provide a more rigorous definition of stability, namely when non-linearity come into play.

Two types of unperturbed motion are particularly worth of notice:

- i) the system being set to a constant position (equilibrium position);
- ii) the system performing a periodic motion

In this lesson, we will only consider the stability of the system when perturbed from an equilibrium position.

If we consider a linear mechanical system subjected to a force field linearly dependent on the system's state, the stability analysis with respect to an unperturbed equilibrium is relatively simple, on account of two reasons: first, the linear system has only one equilibrium position; second, the stability of the system does not depend on the amplitude of the perturbation applied.

In contrast, a system subject to a non-linear force field (the case of interest here) may have multiple equilibrium positions, and the perturbed motion originating from the initial perturbation will have different features depending on the amplitude of the perturbation. Therefore, for a system subjected to a non-linear force field we need to introduce the notion of stability in the sense of Liapunov as follows.

First, we assume the equilibrium position to be corresponding to the zero value of the coordinates. There is no loss of generality in this assumption, since a change of coordinates is always possible to have the equilibrium position satisfying this condition.

Then, we denote by $\underline{z}(t)$ the state vector for the system and by $\underline{z}_0$ the initial state of the system following the perturbation.

- i) we say the system is **stable** in the neighbourhood of the equilibrium position $\underline{x} = \underline{0}$ if for any arbitrary positive value ε a value δ exists so that for any initial state satisfying $\|\underline{z}_0\| < \delta$ the perturbed state remains limited as follows: $\|\underline{z}(t)\| < \varepsilon$ for any time $0 \leq t \leq \infty$;
- ii) the system is **asymptotically stable** if it is stable and if additionally the perturbed state tends to zero with growing time: $\lim_{t \rightarrow \infty} \|\underline{z}(t)\| = 0$
- iii) the system is **unstable** if it is not stable;

The direct use of the above definitions requires that the equations of motion (1.1) or (1.2) are solved considering the effect of the non-linear terms introduced by the force field. This is however generally not possible in analytical form, whereas an approximate numerical solution would not allow to check that the stable behaviour because an infinite number of possible initial perturbations satisfying the amplitude limit set in i) should be analysed.

Therefore a different procedure, based on a theorem due to Lyapunov, needs to be used to analyse the stability of the system. To this aim, we linearize the equations of motion of the system in the neighbourhood of the equilibrium position of interest, and we check the stability of the linearized system, which can be done by simply computing the eigenvalues of the state matrix. Finally, the results of the stability analysis performed for the linearized system are transferred to the non-linear system according to the following rules:

- i) if the linearised system is asymptotically stable, the non-linear system is asymptotically stable (in the sense of Lyapunov);
- ii) if the linearised system is unstable, the non-linear system is unstable;
- iii) if the linearised system is stable but not asymptotically stable, nothing can be concluded about the stability of the non-linear system by means of this analysis, and other methods should be used to analyse the stability of the system.

3 Stability of a 1-d.o.f. system subject to a force field.

In this section, we perform the stability analysis of a single-d.o.f. system subject to a non-linear, autonomous force field, as shown in Figure 3.1. The equation of motion for the system reads:

$$m_s \ddot{x} + c_s \dot{x} + k_s x = F(x, \dot{x}) \quad (3.1)$$

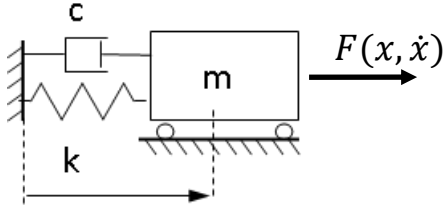


Figure 3.1: Single-d.o.f. system subject to a force field.

The equilibrium positions for the system are found assuming a solution of (3.1) in the form of a constant position x_0 :

$$x = x_e ; \dot{x} = 0 ; \ddot{x} = 0 \quad (3.2)$$

Introducing (3.2) in (3.1) we get:

$$k_s x_e = F(x_e, 0) \quad (3.3)$$

Equation (3.3) is an ordinary, non-linear equation and provides a different number of solutions depending on the specific mathematical expression of the force field. A separate stability analysis needs to be performed for each of these solutions.

To this aim, we linearize the expression of the force field around the equilibrium position of interest:

$$F(x, \dot{x}) = F(x_e, 0) + \left. \frac{\partial F}{\partial x} \right|_e (x - x_e) + \left. \frac{\partial F}{\partial \dot{x}} \right|_e \dot{x} \quad (3.4)$$

Note that in (3.4) the notation $\left|_e\right.$ is used to specify that the partial derivatives of the function defining the force field are evaluated at the equilibrium state $x = x_e, \dot{x} = 0$.

Next, we introduce a field force stiffness k_f and a field force damping c_f as follows:

$$k_f = - \left. \frac{\partial F}{\partial x} \right|_e ; c_f = - \left. \frac{\partial F}{\partial \dot{x}} \right|_e \quad (3.5)$$

and we perform a change of reference replacing the independent coordinate x with the new coordinate $\bar{x}$, so that the new variable has zero value in the equilibrium position for which the stability analysis is being carried out:

$$\bar{x} = x - x_e ; \dot{\bar{x}} = \dot{x} ; \ddot{\bar{x}} = \ddot{x} \quad (3.6)$$

Replacing (3.4), (3.5) and (3.6) in (3.1) and taking into account (3.3) we get:

$$m_s \ddot{\bar{x}} + (c_s + c_f) \dot{\bar{x}} + (k_s + k_f) \bar{x} = 0 \quad (3.7)$$

We see from (3.7) that after linearization the field force acts in the form of adding a stiffness term k_f and a damping term c_f to the structural stiffness and damping parameters. We also note that because the additional stiffness and damping terms are obtained as the gradients of the field force in the equilibrium position, cf. (3.5), these parameters may be positive but also negative valued, whereas the structural terms are assumed to be always positive valued. As a consequence, the overall stiffness k_T and overall damping c_T defined as:

$$k_T = k_s + k_f \quad ; \quad c_T = c_s + c_f \quad (3.8)$$

may be positive or negative valued.

Based on equation (3.8), the stability analysis is performed as follows. First, we assume a solution for (3.7) in the form:

$$\bar{x} = Xe^{\lambda t} \quad ; \quad \dot{\bar{x}} = \lambda Xe^{\lambda t} \quad ; \quad \ddot{\bar{x}} = \lambda^2 Xe^{\lambda t} \quad (3.9)$$

with X and λ two complex-valued constants. Replacing (3.8) and (3.9) in (3.7) we have:

$$(\lambda^2 m_s + \lambda c_T + k_T)Xe^{\lambda t} = 0 \quad (3.10)$$

Non trivial solutions (3.10) are obtained for parameters λ satisfying the following condition:

$$\lambda^2 m_s + \lambda c_T + k_T = 0$$

which gives:

$$\lambda_{1,2} = -\frac{c_T}{2m_s} \pm \sqrt{\left(\frac{c_T}{2m_s}\right)^2 - \frac{k_T}{m_s}} = 0 \quad (3.11)$$

The values of $\lambda_{1,2}$ and hence the outcome of the stability analysis depend on the values of the overall stiffness and damping parameters k_T and c_T . Three cases need to be considered:

Case 1: $k_T > 0$ and $c_T > 0$

In this case, the two solutions $\lambda_{1,2}$ have both negative-valued real parts. Indeed, if:

$$c_T < 2m_s \sqrt{\frac{k_T}{m_s}}$$

the two solutions are complex conjugate, with negative real part (recall that $m_s > 0$). Otherwise, the two solutions are real-valued and, because $k_T > 0$ the first term at second side of (3.11) has a larger absolute value than the square root, so that regardless the sign used for the square root the two values of λ are negative.

In both cases, the linearized system governed by equation (3.7) is asymptotically stable, and so is the complete non-linear system, on account of the Lyapunov's theorem.

Case 2: $k_T < 0$ and $c_T > 0$

In this case, the two solutions $\lambda_{1,2}$ are real-valued, one negative and one positive:

$$\lambda_1 = -\alpha_1$$

$$\lambda_2 = \alpha_2$$

with α_1 and α_2 two real-valued positive constants. The linearized system is in this case unstable, and so is the complete non-linear system. The perturbed motion in the neighbourhood of the equilibrium position is described by:

$$\bar{x} = A_1 e^{-\alpha_1 t} + A_2 e^{\alpha_2 t} \quad (3.12)$$

and corresponds to a time history of the type shown in figure 3.2: for a small initial perturbation (initial position slightly displaced from the equilibrium $\bar{x} = 0$ and/or initial velocity slightly different from zero) the system moves further away from the equilibrium position, without oscillations. This kind of behaviour is called *divergence-type instability*.

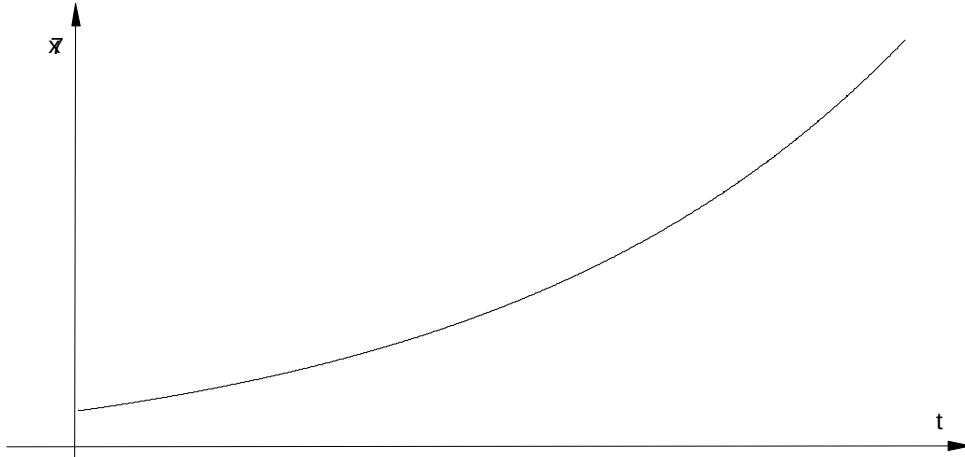


Figure 3.2: Divergence-type instability.

Case 3: $k_T > 0$ and $c_T < 0$

Provided that the absolute value of the total damping parameter is sufficiently small:

$$|c_T| < 2m_s \sqrt{\frac{k_T}{m_s}}$$

the two solutions $\lambda_{1,2}$ are complex-conjugate, with positive real part:

$$\lambda_{1,2} = \alpha \pm i\omega$$

with α and ω real-valued positive constants. The linearized system is in this case unstable, and so is the complete non-linear system. The perturbed motion in the neighbourhood of the equilibrium position is described by:

$$\bar{x} = e^{\alpha t} (A_1 e^{-i\omega t} + A_2 e^{i\omega t}) = e^{\alpha t} [A \cos(\omega t) + B \sin(\omega t)] \quad (3.13)$$

which is an oscillation with increasing amplitude, as shown in figure 3.3. This instability is caused by the negative damping term, which instead of dissipating the energy of vibration inputs mechanical energy in the vibration of the system, so that with the increase of the cycles of oscillation the amplitude of vibration becomes larger. This is called *dynamic instability*.

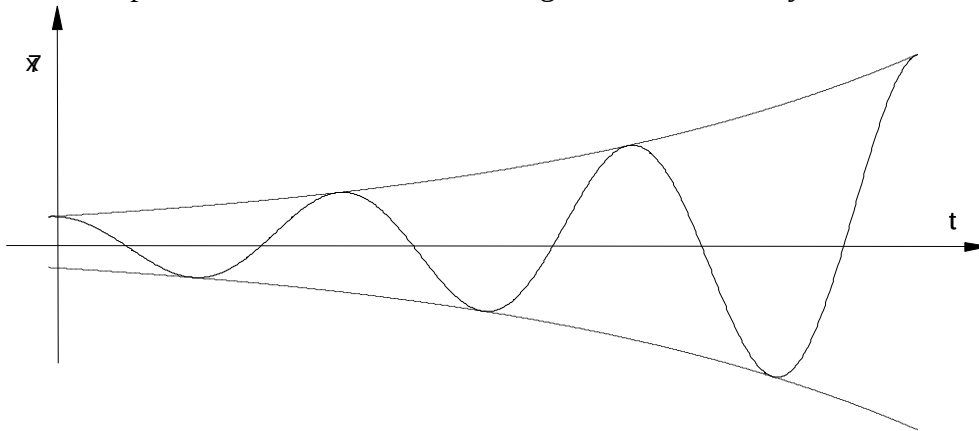


Figure 3.3: Dynamic instability.

Case 4: $k_T < 0$ and $c_T < 0$

In this case the solutions are again real valued, one negative and one. Therefore the system is unstable around the considered equilibrium position and shows a divergence-type instability.

3.1 An example of dynamic instability: friction-induced self-excited vibration

The simple pendulum is an example of a system showing a divergence-type unstable behaviour in the neighbourhood of one equilibrium position (namely, the one in which the centre of mass of the body is set above the hinge defining the centre of rotation for the body).

In this section an example is provided of a 1-d.o.f. system showing a dynamic instability. The system is shown in Figure 3.4 and consists of a 1-d.o.f. vibrating slider placed over a belt conveyor moving with speed V . A vertical load P presses the slider against the conveyor, and we assume dry friction to occur at the interface between the slider and the conveyor.

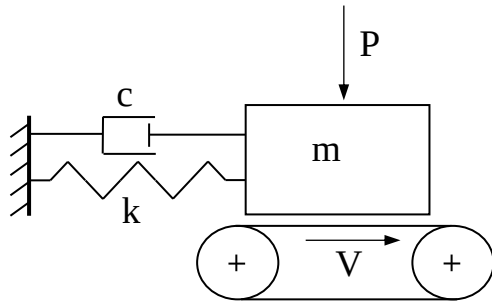


Figure 3.4: Exemplification of a 1-d.o.f. system subject to friction-induced self-excited vibration.

In presence of sliding, the dry friction force T exchanged by the slider with the conveyor is expressed as a function of the normal contact force N and of the sliding speed V_s as:

$$T = f_d(V_s)N \quad (3.14)$$

with f_d the dry friction coefficient which varies with the sliding speed as shown qualitatively in Figure 3.5: the friction coefficient is constant with speed only for relatively large values of the sliding speed, whereas for low sliding speed it is (in first approximation) linearly decreasing with the speed. If we confine our analysis to low sliding speeds, the friction coefficient can be expressed as:

$$f_d = f_{d0} - \alpha V_s \quad (3.15)$$

with f_{d0} and α two constant parameters, representing respectively the value of the friction coefficient for zero sliding speed and the absolute value of the initial slope of the diagram.

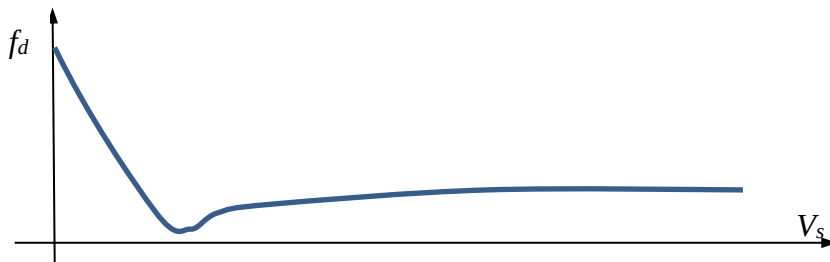


Figure 3.5: Dry friction coefficient as function of the sliding speed

For the case shown in Figure 3.4, the sliding speed is:

$$V_s = V - \dot{x} \quad (3.16)$$

In order to write the equation of motion for the slider, we use D'Alembert's principle. The forces acting on the slider are shown in figure 3.6.

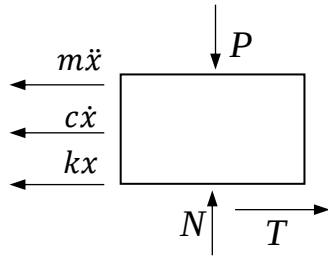


Figure 3.4: Exemplification of a 1-d.o.f. system subject to friction-induced self-excited vibration.

The vertical equilibrium of the slider brings:

$$N - P = 0 \quad (3.17)$$

whereas the horizontal dynamic equilibrium of the slider result in the equation:

$$m\ddot{x} + c\dot{x} + kx = T \quad (3.18)$$

Solving (3.17) for the normal force N and replacing in (3.14), considering (3.15) and (3.16) brings:

$$T = [f_{d0} - \alpha(V - \dot{x})]P \quad (3.19)$$

and finally, replacing (3.19) in (3.18) we obtain:

$$m\ddot{x} + c\dot{x} + kx = (f_{d0} - \alpha V)P + \alpha P\dot{x} \quad (3.20)$$

which can be rewritten as:

$$m\ddot{x} + (c - \alpha P)\dot{x} + kx = (f_{d0} - \alpha V)P \quad (3.21)$$

if the structural damping c of the system is too low, the total damping of the system, including the negative term caused by the dry friction force field becomes negative and is cause of a dynamic instability known as friction-induced self-excited vibration. The effects of this phenomenon can be easily experienced in common life: the vibration of a violin chord is amplified by the slow sliding of the bow and the squeal noise heard from a car braking can be described in simplified terms according to the model introduced above.

4 Stability of a 2-d.o.f. system subject to a force field.

In this section, we perform the stability analysis for a 2-d.o.f. system. The concepts introduced here can be extrapolated to the case of a generic multi-d.o.f. system. The mechanical system we consider is show in figure 4.1 and consists of a point mass moving in a plane subject to stiffness and damping in the two directions of a Cartesian reference x - y .

$$\begin{aligned} m_s \ddot{x} + c_{sx} \dot{x} + k_{sx} x &= F_x(x, \dot{x}, y, \dot{y}) \\ m_s \ddot{y} + c_{sy} \dot{y} + k_{sy} y &= F_y(x, \dot{x}, y, \dot{y}) \end{aligned} \quad (4.1)$$

which can be rewritten in matrix form as:

$$[M_s] \ddot{\underline{x}} + [C_s] \dot{\underline{x}} + [K_s] \underline{x} = \underline{F}(\underline{x}, \dot{\underline{x}}) \quad (4.2)$$

with:

$$\underline{\dot{x}} = \begin{Bmatrix} \dot{x} \\ \dot{y} \end{Bmatrix} ; [M_s] = \begin{bmatrix} m_s & 0 \\ 0 & m_s \end{bmatrix} ; [C_s] = \begin{bmatrix} c_{sx} & 0 \\ 0 & c_{sy} \end{bmatrix} ; [K_s] = \begin{bmatrix} k_{sx} & 0 \\ 0 & k_{sy} \end{bmatrix} ; \underline{F} = \begin{Bmatrix} F_x \\ F_y \end{Bmatrix} \quad (4.3)$$

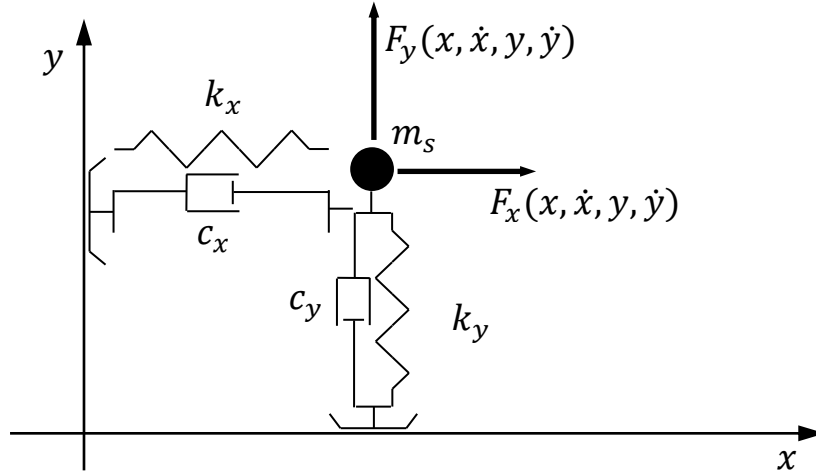


Figure 4.1: Example of 2-d.o.f. system subject to a force field.

The equilibrium positions of the system are defined by solutions in the form:

$$\underline{x} = \underline{x}_e ; \underline{\dot{x}} = \underline{0} ; \underline{\ddot{x}} = \underline{0} \quad (4.4)$$

Introducing (3.2) in (3.1) we get:

$$[K_s] \underline{x}_e = \underline{F}(\underline{x}_e, \underline{0}) \quad (4.5)$$

Equation (4.5) is equivalent to a system of non-linear ordinary equations and generally need to be solved using numerical techniques. The stability analysis needs then to be performed separately for each equilibrium position of interest.

Like in the case of the 1-d.o.f. system, as a first step we perform the linearization of the force field around the equilibrium position:

$$\underline{F}(\underline{x}, \underline{\dot{x}}) = \underline{F}(\underline{x}_e, \underline{0}) + \left. \frac{\partial \underline{F}}{\partial \underline{x}} \right|_e (\underline{x} - \underline{x}_e) + \left. \frac{\partial \underline{F}}{\partial \underline{\dot{x}}} \right|_e \underline{\dot{x}} \quad (4.6)$$

In (4.6) $|_0$ denotes that the partial derivatives of the force field are evaluated in the equilibrium position and for zero velocity.

Then, we introduce the field force stiffness matrix $[K_F]$ and the field force damping matrix $[C_F]$ defined as:

$$[K_F] = - \left. \frac{\partial \underline{F}}{\partial \underline{x}} \right|_e = \begin{bmatrix} - \left. \frac{\partial F_x}{\partial x} \right|_e & - \left. \frac{\partial F_x}{\partial y} \right|_e \\ - \left. \frac{\partial F_y}{\partial x} \right|_e & - \left. \frac{\partial F_y}{\partial y} \right|_e \end{bmatrix} ; [C_F] = - \left. \frac{\partial \underline{F}}{\partial \underline{\dot{x}}} \right|_e = \begin{bmatrix} - \left. \frac{\partial F_x}{\partial \dot{x}} \right|_e & - \left. \frac{\partial F_x}{\partial \dot{y}} \right|_e \\ - \left. \frac{\partial F_y}{\partial \dot{x}} \right|_e & - \left. \frac{\partial F_y}{\partial \dot{y}} \right|_e \end{bmatrix} \quad (4.7)$$

Finally, we perform the change of reference:

$$\bar{x} = x - x_e ; \quad \dot{\bar{x}} = \dot{x} ; \quad \ddot{\bar{x}} = \ddot{x} \quad (4.8)$$

Replacing (4.6), (4.7) and (4.8) in (4.2) and taking into account (4.5) we get:

$$[M_s] \ddot{\bar{x}} + ([C_s] + [C_F]) \dot{\bar{x}} + ([K_s] + [K_F]) \bar{x} = 0 \quad (4.9)$$

or:

$$[M_s] \ddot{\bar{x}} + [C_T] \dot{\bar{x}} + [K_T] \bar{x} = 0 \quad (4.10)$$

with:

$$[C_T] = [C_s] + [C_F] ; \quad [K_T] = [K_s] + [K_F] \quad (4.11)$$

It is easy to generalise the results above, so that for a generic n -d.o.f. system subject to a field force the non-linear equation of motion will be in the form (4.2) with matrices having dimension n instead of 2. In the same way, the linearized equation for the system will be in the form (4.9) or (4.10), with matrices $[C_F]$ and $[K_F]$ defined according to a straightforward generalisation of (4.7). The stability analysis for the linearized system governed by equation (4.10) is performed as usual assuming the solution:

$$\bar{x} = \underline{X} e^{\lambda t} ; \quad \dot{\bar{x}} = \lambda \underline{X} e^{\lambda t} ; \quad \ddot{\bar{x}} = \lambda^2 \underline{X} e^{\lambda t} \quad (4.12)$$

with $\underline{X}$ and λ two complex-valued constants. Replacing (4.12) in (4.10) we get:

$$(\lambda^2 [M_s] + \lambda [C_T] + [K_T]) \underline{X} e^{\lambda t} = 0 \quad (4.13)$$

This results in a linear, homogenous system, for which non-trivial solutions are obtained for λ satisfying the condition:

$$\det(\lambda^2 [M_s] + \lambda [C_T] + [K_T]) = 0 \quad (4.14)$$

For the 2-d.o.f. system considered here, equation (4.14) results in a 4-th order polynomial equation, whereas for a generic n -d.o.f. system, (4.14) results in a $2 \times n$ -th order polynomial equation. Even in the simplest case of the 2-d.o.f. system, equation (4.14) normally cannot be solved analytically, and needs instead to be solved using appropriate numerical techniques, the simplest consisting of re-writing (4.10) in state-form and computing the solutions λ of (4.14) as the eigenvalues of the state matrix for the linearized system.

Once the solutions of (4.14) are known, the stability / asymptotic stability / instability of the linearized system can be ascertained and, based on the Lyapunov's theorem, the result can be extended to the behaviour of the non-linear system in the neighbourhood of the considered equilibrium position, except in the inconclusive case of the linearized system being stable but not asymptotically stable.

4.1 Stability analysis for the 2-d.o.f. system subject to a positional force field

We consider here the special case when the system shown in figure 4.1 is not subject to structural damping ($[C_s] = 0$) and the force field is only function of the position of the system, but is not affected by the velocity of the system. In mathematical terms:

$$\underline{F} = \underline{F}(x) \quad (4.15)$$

In this case, on account of (4.7) $[C_F] = 0$ and the equation of motion of the linearized system becomes:

$$[M_s] \ddot{\bar{x}} + [K_T] \bar{x} = 0 \quad (4.10b)$$

with $[K_F]$ defined by (4.7). Equation (4.14) then reduces to:

$$\det(\lambda^2 [M_s] + [K_T]) = 0 \quad (4.14b)$$

and, considering the expressions of $[M_s]$ and $[K_T]$ provided by (4.3) and (4.11), equation (4.14b) becomes:

$$\lambda^4 m_s^2 + \lambda^2 (K_{Txx} + K_{Tyy}) m_s + (K_{Txx} K_{Tyy} - K_{Txy} K_{Tyx}) = 0 \quad (4.14c)$$

As a particular case of (4.14), this is a biquadratic equation that can be easily solved. To this aim, we first transform (4.14c) by dividing both terms by m_s^2 and we define:

$$\bar{K}_{Txx} = \frac{K_{Txx}}{m_s} ; \bar{K}_{Txy} = \frac{K_{Txy}}{m_s} ; \bar{K}_{Tyx} = \frac{K_{Tyx}}{m_s} ; \bar{K}_{Tyy} = \frac{K_{Tyy}}{m_s}$$

so that (4.14c) becomes:

$$\lambda^4 + \lambda^2 (\bar{K}_{Txx} + \bar{K}_{Tyy}) + (\bar{K}_{Txx} \bar{K}_{Tyy} - \bar{K}_{Txy} \bar{K}_{Tyx}) = 0 \quad (4.14d)$$

The solutions λ of (4.14d) are provided by the square roots of:

$$\lambda^2 = -\frac{\bar{K}_{Txx} + \bar{K}_{Tyy}}{2} \pm \sqrt{\left(\frac{\bar{K}_{Txx} + \bar{K}_{Tyy}}{2}\right)^2 - (\bar{K}_{Txx} \bar{K}_{Tyy} - \bar{K}_{Txy} \bar{K}_{Tyx})} \quad (4.16)$$

which can be rewritten as:

$$\lambda^2 = -\frac{\bar{K}_{Txx} + \bar{K}_{Tyy}}{2} \pm \sqrt{\left(\frac{\bar{K}_{Txx} - \bar{K}_{Tyy}}{2}\right)^2 + \bar{K}_{Txy} \bar{K}_{Tyx}} \quad (4.16b)$$

Before we use equations (4.16) and (4.16b) to perform the stability analysis for the 2-d.o.f. system subjected to a positional force field, we recall what is a *conservative* force field.

Definition and properties of a conservative force field

We consider below the case of a positional force field acting in the x-y plane, but the generalisation to the three-dimensional case is straightforward.

Given a point P moving in the x-y plane and subject to the force field F , cf. Figure 4.2, the work performed by the force field when point P moves from an initial position A to a final position B along a given path Γ is defined as the line integral:

$$W = \int_{\Gamma} [F_x(x, y) dx + F_y(x, y) dy] \quad (4.17)$$

The force field is said to be *conservative* if the work W defined by (4.17) does not depend on the particular path followed by point P , but only on the position of start and end points A and B . If this property holds, it means a function V can be defined so that:

$$W = \int_{\Gamma} [F_x(x, y) dx + F_y(x, y) dy] = V(x_A, y_A) - V(x_B, y_B) \quad (4.18)$$

the function V is called the *potential energy* associated with the conservative force field.

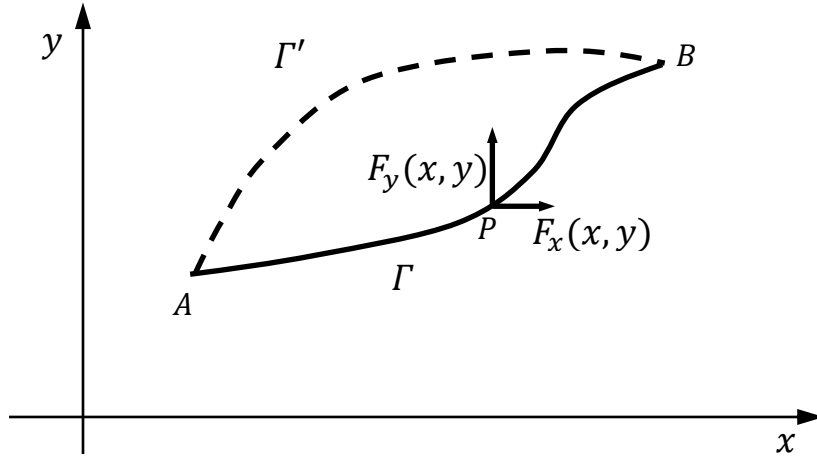


Figure 4.2: Work of the positional force field for point P moving from P' to P''.

Applying (4.18) to the displacement of the point from $P(x, y)$ to $P+dP$ we get:

$$dW = F_x(x, y)dx + F_y(x, y)dy = -dV = -\frac{\partial V}{\partial x}dx - \frac{\partial V}{\partial y}dy \quad (4.19)$$

so, we conclude that the following relationships hold between the partial derivatives of the potential energy and the Cartesian components of the force field:

$$F_x = -\frac{\partial V}{\partial x} ; F_y = -\frac{\partial V}{\partial y} \quad (4.20)$$

or, equivalently, the vector defining the force field is the opposite of the gradient of the potential energy:

$$\vec{F} = -\text{grad}(V) \quad (4.21)$$

Equation (4.20) provides a rule to assess whether or not a force field defined by the scalar components $F_x(x, y)$, $F_y(x, y)$ is conservative. Indeed, the potential energy function V , if existing, must satisfy the Schwarz's theorem on second order derivatives of a function of multiple variables, i.e.:

$$\frac{\partial^2 V}{\partial x \partial y} = \frac{\partial^2 V}{\partial y \partial x} \quad (4.22)$$

hence, the force field is conservative if and only if it satisfies the following condition:

$$\frac{\partial F_x}{\partial y} = \frac{\partial F_y}{\partial x} \quad (4.23)$$

Stability analysis for the 2-d.o.f. system

Depending on the properties of the positional force field, different cases may occur.

Case 1: Conservative force field.

In this case, on account of (4.23) and (4.7), $\bar{K}_{Txy} = \bar{K}_{Tyx}$. Therefore, based on (4.16b), the squares of the solutions λ are real valued. We need to further distinguish two sub-cases.

Case 1a: Conservative force field, positive definite matrix $[K_T]$.

Based on (4.16), in this case the squares of the solutions λ are negative, because the trace and the determinant of matrix $[K_T]$ are both real and positive valued. Taking the square root, the solutions are found in the form of two purely imaginary complex conjugate couples:

$$\lambda_{1,2} = \pm i\omega_1$$

$$\lambda_{3,4} = \pm i\omega_2$$

with ω_1 and ω_2 real positive values.

It can therefore be concluded that the linearized system is stable, but not asymptotically stable: this is the non-conclusive case of the Lyapunov's theorem, and therefore the stability of the system needs to be assessed by a different means.

In this case, a theorem by Dirichlet can be applied, which states the following: for a system subject only to conservative forces, the equilibrium positions correspond to stationary potential energy, and the stable equilibrium positions correspond to a minimum of the potential energy. If matrix $[K_T]$ is positive definite, it means:

$$V = \frac{1}{2} \bar{x}^T [K_T] \bar{x} > 0 \quad \forall \bar{x} \neq \underline{0}$$

the above property of V shows that the potential energy is increased for any (sufficiently small) displacement of the system from the equilibrium position $\bar{x} = \underline{0}$, which means it is minimum in the equilibrium position. Hence, it can be concluded that the equilibrium position is stable. More generally, an n -d.o.f. system having a positive definite matrix $[K_T]$ will be stable (non asymptotically).

Case 1b: Conservative force field, non-positive definite matrix $[K_T]$.

In this case, either the determinant or the trace of matrix $[K_T]$ (or both) are negative. Therefore, based on (4.16), at least one of the squares of the solutions λ is real and positive valued. The solutions λ then are either in the form:

$$\lambda_{1,2} = \pm i\omega$$

$$\lambda_{3,4} = \pm \alpha$$

with α and ω real positive values, or in the form:

$$\lambda_{1,2} = \pm \alpha_1$$

$$\lambda_{3,4} = \pm \alpha_2$$

with α_1 and α_2 real positive values. In both case, at least one of the roots λ is real valued and positive and hence the linearized system is unstable. Based on Lyapunov's theorem, it is then concluded that the system is unstable in the neighbourhood of the considered equilibrium position and shows a divergence-type instability. More generally, an n -d.o.f. system having non-positive definite matrix $[K_T]$ will show a divergence-type instability.

Case 2: Non-conservative force field.

In this case, $\bar{K}_{Txy} \neq \bar{K}_{Tyx}$. We are particularly interested in the case when the following conditions apply:

$$\bar{K}_{Txy} \bar{K}_{Tyx} < 0 \quad \text{and} \quad \left| \bar{K}_{Txy} \bar{K}_{Tyx} \right| > \left(\frac{\bar{K}_{Txx} - \bar{K}_{Tyy}}{2} \right)^2 \quad (4.24)$$

If the above conditions apply, based on (4.16b) the squares of the solutions λ are obtained in the form of a complex conjugate pair:

$$\lambda_{I,II}^2 = M e^{\pm i\psi}$$

The solutions are hence:

$$\lambda_1 = \sqrt{M} e^{i\frac{\psi}{2}} ; \lambda_2 = \sqrt{M} e^{i\left(\frac{\psi}{2} + \pi\right)} ; \lambda_3 = \sqrt{M} e^{-i\frac{\psi}{2}} ; \lambda_4 = \sqrt{M} e^{-i\left(\frac{\psi}{2} + \pi\right)} \quad (4.25)$$

As shown in Figure 4.3, the solutions are two pairs of complex conjugate values, one couple having positive real part. It is then concluded that in the neighbourhood of the considered equilibrium position the system shows a dynamically unstable behaviour, i.e. oscillations with amplitude increasing with time.

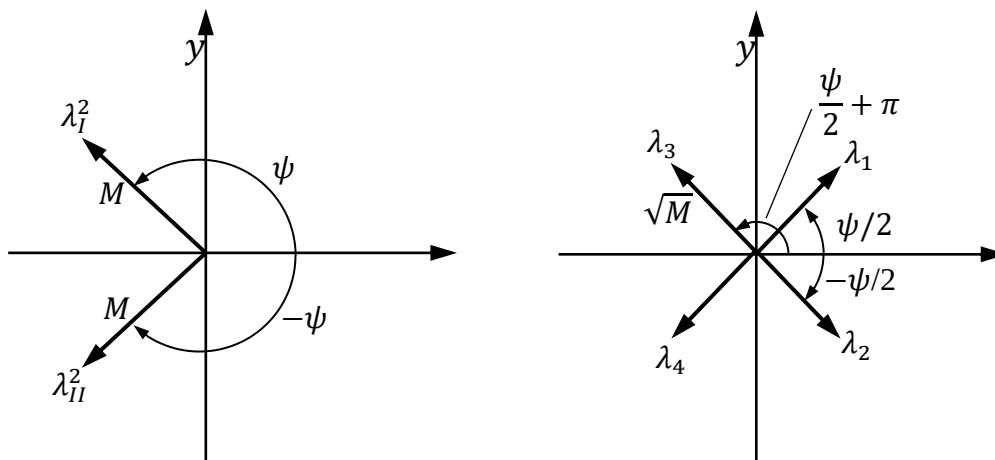


Figure 4.3: Roots of the system in case of ‘flutter instability’.

Differently from the 1-d.o.f. system treated in Section 3, this instability is not due to a ‘negative damping’ effect generated by the field force. Instead, the increase in the energy of vibration is caused by the work of the non-conservative force field. It can be shown (cf. Diana and Cheli for details) that the two complex modes of vibration associated with solutions (4.25) represent the same elliptic trajectory of the mass in the x - y plane, described in clockwise and counter-clockwise direction.

Recalling that the work of a non-conservative force field is not equal to zero when point P describes a closed path, and noting that the sign of the work changes when the direction of the point describing the closed path changes, it is concluded that one of the two modes of vibration corresponds to negative work performed by the force field (thereby resulting in a damped oscillation), but the other mode of vibration corresponds to the same trajectory negotiated in the opposite direction which corresponds to positive work of the force field and therefore in increasing amplitude of vibration.

In conclusion, the dynamic instability resulting from a non-conservative force field satisfying conditions (4.24) is caused by the energy introduced in the vibration by the non-conservative force field. This type of instability is called ‘flutter instability’, this name comes from the observation of ‘fluttering’ in airplane wings as the result of aero-elastic instability leading to self-excited vibrations.